Daniel M. Cherenson

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Summary

Third-year Robotics Ph.D. candidate with 3 years of aerospace industry experience as a guidance, navigation, and control (GNC) engineer. My research focuses on safety-critical planning and control under uncertainty, with emphasis on robust safety, uncertainty quantification, and online adaptation.

Education

Ph.D. in Robotics 2024 - est. 2028
University of Michigan
Advised by Dimitra Panagou in the Distributed Autonomous Systems and Control Lab; GPA: 4.00
Awards: 2024 NSF GRFP Honorable Mention, Departmental First Year Fellowship

M.S. in Electrical Engineering 2021
Washington University in St. Louis
GPA: 4.00; Bachelor's/Master's Scholarship

B.S. in Mechanical Engineering 2017 - 2021
Washington University in St. Louis
GPA: 3.97; Summa Cum Laude

Research Areas and Technical Expertise

Safety-Critical Control, Robust & Adaptive Control, Guidance & Motion Planning, Optimal Control, Multi-Robot Systems, State Estimation, Uncertainty Quantification, Learning & Optimization for Control and Robotics

Professional Experience

GNC Engineer March 2024 - July 2024
Near Earth Autonomy *Pittsburgh, PA*

- Operated the autonomy payload and directed integrated flight tests of the L3Harris FVR-90, successfully verifying LiDAR and computer vision performance ahead of demo at RIMPAC 2024 [Video]
- Developed an IMU and error state Kalman filter (ESKF) simulation in MATLAB and evaluated performance of various IMUs for visual-inertial odometry in GPS-denied environments

GNC Engineer III January 2023 - February 2024
L3Harris Technologies (formerly Latitude Engineering) *Tucson, AZ*

- Developed, tested, and maintained GNC algorithms, simulation sofware, embedded software, and application software for the FVR-90 hybrid quadrotor-fixed-wing VTOL UAV
- Designed guidance and control laws using first principles and implemented flight code in C, improving performance during highly nonlinear landing transition flight phases
- Designed and tested embedded and application software in C++ for interfacing with Near Earth Autonomy's computer vision and LiDAR payload for safe autonomous landing on ships and in cluttered environments
- Conducted software- and hardware-in-the-loop testing and collaborated with flight operations team to plan flight tests
- Implemented on-board diagnostic test to abort landings with failed start-ups of VTOL motors, avoiding over $2 million in potential aircraft losses
- Analyzed flight telemetry and presented incident report to foreign military customer, resulting in increased trust and renewed multimillion-dollar contract

- Led integration of an L3Harris legacy airplane autopilot with our VTOL flight code in Simulink and C++

GNC Engineer II June 2021 - January 2023
Raytheon Missiles and Defense *Tucson, AZ*

- Named as a subject matter expert on terminal guidance algorithms for a major naval missile program
- Developed and integrated an optimal guidance law that accounts for higher-order missile dynamics to improve performance in low dynamic pressure intercepts
- Conducted terminal guidance trade studies on interacting multiple model filters and optimal guidance; implemented these methods which improved missile performance metrics by up to 10%
- Developed and maintained a MATLAB simulation for hypersonic boost-glide trajectory generation, adopted for target simulation by multiple programs within the company
- Developed C++ and Python simulation code for a large-scale global sensor network for an IRAD program developing a distributed consensus assignment and tracking algorithm
- Presented flight test analysis to government customers at a missile data review

Publications

Journal Publications

1. **D. M. Cherenson***, H. Lee*, T. Kim, and D. Panagou, "Distributionally Robust Safety Under Arbitrary Uncertainties: A Safety Filtering Approach," *IEEE Robotics and Automation Letters*, 2026. [Project] [Demo]

Peer-Reviewed Conference Publications

1. **D. M. Cherenson** and D. Panagou, "Staggered Integral Online Conformal Prediction for Safe Dynamics Adaptation with Multi-Step Coverage Guarantees," *IEEE Conf. on Decision and Control (CDC)*, 2026. [Code]
2. T. M. Vielmetti, **D. M. Cherenson**, and D. Panagou, "Multi-Objective Kinodynamic Planning in Stochastic Environments," *IEEE Conference on Decision and Control (CDC)*, 2026.
3. **D. M. Cherenson**, D. R. Agrawal, and D. Panagou, "Autonomy Architectures for Safe Planning in Unknown Environments Under Budget Constraints," *American Control Conference (ACC)*, 2026. [Code] [Demo]
4. **D. M. Cherenson** and D. Panagou, "Adaptive Control Allocation for Underactuated Time-Scale Separated Non-Affine Systems," *American Control Conference (ACC)*, 2026. [Code]

Skills

Programming: Python, C/C++, Julia, MATLAB/Simulink
Libraries: JAX, PyTorch, OpenCV, CVXPY, Acados, MATLAB Optimization and Control System Toolboxes
Tools: Git, Linux, ROS/ROS 2, Docker
Languages: English (native), French (professional working proficiency)

Leadership Experience

Master's Student Advisor September 2026 - present
University of Michigan, ROB 590 Directed Study

- Proposed a semester-long directed study for supervision and mentorship of a Master's student, teaching optimization, multi-robot systems, safety-critical control, adaptive control, and general research skills

Graduate Student Instructor September 2025 - December 2025
University of Michigan, ROB 501 Mathematics for Robotics

- Gave weekly recitation lectures and held office hours for a graduate-level math course of 102 students

Service

Alumni Mentor August 2023 - present
Washington University McKelvey School of Engineering